

Multiple Choice (10 marks)

1. A surgical support can be used for:
 - (a) retraction of the stoma.
 - (b) high output stomas.
 - (c) herniation around the stoma.
 - (d) infection around the stoma.
 - (e) dehydration caused by the stoma.
2. All other things being equal, which of the following persons is likely to live the longest?
 - (a) A black woman
 - (b) A Hispanic man
 - (c) A white woman
 - (d) An Asian man
 - (e) A white man
3. What must you ensure before a speaking valve is used?
 - (a) The patient has no difficulty in breathing.
 - (b) It is 24 hours after the tracheostomy tube was first inserted.
 - (c) The patient is able to speak clearly.
 - (d) The patient's vital signs are stable.
 - (e) That the patient is not on a ventilator.
4. The energy released from the breakdown of the high-energy phosphates, ATP and phosphocreatine, can sustain maximal exertion exercise for about:
 - (a) 20-30 minutes.
 - (b) 1-2 seconds.
 - (c) 10-15 minutes.
 - (d) 30-40 seconds.
 - (e) 90-100 seconds.
5. Which of the following is not a beneficial effect of exercise in people with diabetes:
 - (a) Decreased muscle mass
 - (b) Impaired digestion
 - (c) Worsening of vision
 - (d) Increase in blood sugar levels
 - (e) Reduction of bone density

True / False (10 marks)

Answer True or False

6. True or False: The descending loop of Henle is permeable to H^+ , allowing for the reabsorption of this ion in the nephron.
7. True or False: Cognitive symptoms are less likely to manifest in Parkinson's disease compared to other forms of dementia.
8. True or False: Research suggests that women who have fewer children often experience greater longevity than those with larger families.
9. True or False: Vitamin D acts as a cofactor for hydroxylation reactions involved in collagen synthesis.
10. True or False: A baby with pulmonary hypoplasia from oligohydramnios is classified as having a syndrome related to renal agenesis.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of ectomesenchyme in the development of various structures in the body, specifically focusing on melanocytes and their significance in health sciences.
12. Discuss the implications of the increasing age for receiving full social security benefits on public health and individual well-being in contemporary society.

End of Paper